Part - A

$$\begin{bmatrix} 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \end{bmatrix}$$ 5×5 grey image

Sobel - X Kernel

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

1. Dimension of output

$\Rightarrow n - k + 1$

$n = 5$

$k = 3$

$\therefore$ Dim $= 5 - 3 + 1 = \underline{\underline{3}}$

$\therefore$ output $= 3 \times 3$ matrix

2. $$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix} * \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -10 & 0 & 10 \\ -20 & 0 & 20 \\ -10 & 0 & 10 \end{bmatrix} \Rightarrow \underline{\underline{0}} = x_1$$

$$\begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -10 & 0 & 80 \\ -20 & 0 & 160 \\ -10 & 0 & 80 \end{bmatrix}$$

$$\Rightarrow 280 - x_2$$

$$\begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -10 & 0 & 80 \\ -20 & 0 & 160 \\ -10 & 0 & 80 \end{bmatrix}$$

$$\Rightarrow 280 \Rightarrow x_3$$

$x_4 = x_1 = x_7$

$x_5 = x_2 = x_8$

$x_6 = x_3 = x_9$

$$\therefore \; x \text{ Sobel} = \begin{bmatrix} 0 & 280 & 280 \\ 0 & 280 & 280 \\ 0 & 280 & 280 \end{bmatrix}$$

$3 \times 3$

In the input matrix the edge is in between C3 & C4 whereas in x Sobel matrix the change is in between $C_1$ & $C_2$

The sobel operator uses elementwise matrix multiplication and then summation. The sobel operator's coeff perfectly cancel out each other if the multiplied quantity is same or homogeneous.